Oscar I. Hernandez

Curriculum Vitæ

Email: 98Hernandez.Oscar@gmail.com Tel: +1 (646) 961-3715

EDUCATION

Bard College at Simon's Rock (BCSR), Great Barrington, MA
Bachelor of Arts, Mathematics and Computer Science, May 2017
Associate in Arts, May 2015

EXPERIENCE

Entrepreneurship

Automation Engineer, Software Lead Self
January 2018 – Riverside, CA

- Design high-throughput hardware systems to automate additive manufacturing facilities, using a combination of off-the-shelf and custom components as needed.
- Write controls/feedback software systems, including back-end database API and front-end monitor display.
- Administrate company's personal computers, microcomputers, microcontrollers, servers, and their interfaces.
- Write full-stack web applications for various commercial applications, such as syncing data cross e-commerce platforms, file hosting services, and sending data via cloud services.
- Design functional attractive web pages for clients in the retail, gaming, and blogging industries.

Teaching

5th Grade Associate STEM Teacher Success Academy Charter Schools
July, 2017 – December, 2017 New York, NY

- Trained to teach 5th Grade Mathematics and Earth Science at a fast-paced public charter middle school in Harlem, New York.
- Served as the school's sole computer science teacher for grades 5-8, designing, implementing, and teaching it from scratch.
- Co-taught 5th Grade Humanities, serving as the lead History teacher and assistant ELA teacher.

Subject and Class Tutor BCSR
Fall 2014 – Spring 2017 GB, MA

- College: Accelerated Beginning Latin I, Introduction to Computer Science, Formal Logic, Vector Calculus, Algorithms & Data Structures, Discrete Mathematics, and Graph Theory & Combinatorics
- High School: Algebra I with Geometry, Critical Reasoning, Computer Science, Algebra II & Trigonometry

Researh

Research Assistant BCSR
Summer 2015 – Spring 2017 GB, MA

- Explored[Her+18] the computational complexity of the turnstile mechanic in classic games, proving that its inclusion makes games PSPACE-hard in the general case and NP-hard in several limited cases.
- Published[Has+17] findings on dynamics of neural networks, by using my self-written simulation software to predict the behavior of analog neural networks.
- Explored equlibria of trade systems with varying numbers of traders, copies of goods, preferences and wealths.
- Published novel algorithms for constructing binary[DHW16] and k-ary[Dra+18] de Bruijn sequences.

PROJECTS

Publications

[Dra+18] Patrick Dragon et al. "Constructing de Bruijn sequences with Co-Lexicographic Order: The k-ary Grandmama Sequence". In: *European Journal of Combinatorics* 72 (Aug. 2018), pp. 1–11. URL: https://www.sciencedirect.com/science/article/pii/S0195669818300696.

[HW17] Oscar Hernandez and Aaron Williams. "Constructing Maximum-Rotating de Bruijn Sequences Using Necklace Trees". thesis. Great Barrington, MA: Bard College at Simon's Rock, May 2017. URL: http://digitalcommons.bard.edu/sr-theses/1120/.

[Has+17] Harold Hastings et al. "Dynamics of Biomimetic Electronic Artificial Neural Networks". In: *Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics*. Ed. by Visarath In, Patrick Longhini, and Antonio Palacios. Vol. 6. Lecture Notes in Networks and Systems. Denver, CO: Springer International Publishing AG, Mar. 2017, pp. 195–207. URL: https://link.springer.com/chapter/10.1007/978-3-319-52621-8_18.

[DHW16] Patrick Dragon, Oscar Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings". In: *LATIN 2016: Theoretical Informatics*. Ed. by Evangelos Kranakis, Gonzalo Navarro, and Edgar Chávez. Vol. 9644. Lecture Notes in Computer Science. Ensenada, Mexico: Springer-Verlag Berlin Heidelberg, Apr. 2016, pp. 347–361. URL: https://link.springer.com/chapter/10.1007/978-3-662-49529-2_26.

Manuscripts

[Her+18] O. Hernandez et al. "Spinning Around and Around: Turnstile Puzzles are PSPACE-Complete". Apr. 2018.

COMPUTER SKILLS

General: Python, JavaScript, SQL, Shell, Ruby, Haskell, Java, C, C++, C#;
Design: HTML, CSS, jQuery, PHP, Markdown, LaTeX, Blender, Maya, Unity3D;
Applications: Vim, Git+GitHub, DigitalOcean, Heroku, Octave, LibreOffice;
Operating Systems: Arch Linux, Fedora, Ubuntu, Windows, macOS, Android;
Hardware: Raspberry Pi, Arduino, Motors, LEDs.

INTERESTS

Robotic Process Automation, 3D Design/Printing, Game Development, Judaism; Perl, Google Apps Script, Browser Extensions, Alexa Skills, Native Mobile Apps.